

SEAMAPD21: a large-scale reef fish dataset for fine-grained categorization

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Abstract

We present the first large-scale, fine-grained reef fish dataset from the Gulf of Mexico - the Southeast Area Monitoring and Assessment Program Dataset 2021 (SEAMAPD21). Automated image analysis of reef fish species in challenging environments, such as coastal Gulf of Mexico, are crucial for accurate fishery monitoring and stock assessments. As advanced computer vision techniques evolve in precision and capacity, previously difficult tasks such as fish species identification and count due to strong intra-species similarities and inter-species variations, have become feasible. SEAMAPD21 was created to enable fishery scientists and researchers to process raw image-based data at high volumes at a much quicker rate than current manual methods. This open-source dataset contains 90,000 annotations for 130 species of fish from two annual surveys done in 2018 and 2019 in strategic areas of the Gulf of Mexico using baited remote underwater video systems to assess reef fish populations. We perform baseline experiments and document predictive classification results. We document this work at <https://github.com/SEFSC/SEAMAPD21>.

1. Introduction

Fisheries are a multi-billion dollar global industry that require robust management tactics for long-term sustainability. Stock assessments are the back-bone of successful

science-based fisheries management and is a domain that could benefit greatly by leveraging recent advancements made by the computer vision community. The use of in-situ camera systems for monitoring fish abundances have become a common practice in conservation ecology and stock assessment surveys [13], [38], [21]. Deep learning has the incredible potential to facilitate the automated analysis of collected video and images, a step that is usually a bottleneck when these data must be analyzed manually by marine biologists and fishery scientists.

The primary objective of the annual Southeast Area Monitoring and Assessment Program (SEAMAP), from which this dataset is derived, is to provide an index of the relative abundances of fish species associated with topographic features (e.g reefs, banks, and ledges) located on the continental shelf of the Gulf of Mexico (GoM) from Brownsville, TX, USA to the Dry Tortugas, FL, USA (Figure 2). Fishery-independent surveys aid in species mapping, which is an essential tool for both fisheries and conservation programs alike as it provides clear pictures of the distribution of marine resources [28]. Conducting a fishery independent survey of fish associated with the reefs of the northern GoM is difficult because they are distributed over a wide spatial domain stretching from the Florida Keys to Brownsville, TX [33], [14], [5]. This wide terrain is also very diverse: ranging from flat pavement to steep rocky ledges to rugged coral slopes, and occurs in water depths ranging from 15 m to 200 m. Reef fishes are difficult to sample due to habitat complexity, fish species diversity and

mobility of the fish. Additionally, the cryptic nature and behavioral differences among species further compounds problems of estimating reef fish abundance [6], [4]. The survey is conducted on topographic features and therefore, the species targeted are typically classified as reef fish (e.g., red snapper, *Lutjanus campechanus*), but occasionally fish more commonly associated with pelagic environments are observed (e.g. Amberjack, *Seriola dumerili*).

1.1. Related Work

Dataset annotation is a critical part of a supervised detection and classification process. However, collecting and annotating large amounts of data is time-consuming and costly, often becoming the bottleneck when training and deploying models. At the time this paper was written, there exist eight datasets available for fish detection and species identification. These open-sourced datasets are from sites around the United Kingdom [3], [19], Croatia [18], Pakistan [30], and Australia [10], [11], [1], [34]. Our reef fish dataset is the largest available open dataset for fish detection both in the number of annotations and number of species, which is 90000 and 130, respectively. The dataset is captured using a full-spherical camera system at 10 fps that provides 360° field of view by stitching imagery from five cameras and stereo-vision in a single orientation. Most of the datasets provide monocular images, but our dataset includes stereo-vision, which is important for depth estimation for those who want to do 3D object detection.

Historically, marine species detection and identification have been achieved by classical machine learning techniques using handcrafted features such as scale-invariant feature extraction (SIFT) [24]. With recent advances in the field of object detection, models are now capable of multi-object detection and tracking. However, the field of marine science has not yet seen the fruits of these advances, as the standard datasets, and therefore models trained on those datasets, generally target animals such as dogs and cats[9]. While transfer learning can be helpful by pre-training a network’s feature extractors on large scale datasets such as ImageNet and COCO, the resulting models are not designed to consider the fine details of fish such as inter-species features, textures, fin shapes and other distinctive characteristics[20], [36]. Among the common models used for fish detection is CNN-based custom models [29],[37],[26], and [36]. Other research leverages use R-CNN [20] and YOLO [7],[25] as detection models. For in-situ applications, these models must have high accuracy and processing capacity in order to deploy real-time detection, species identification and population count. Among these models, YOLO has been leveraged on GPU based embedded AI computing devices, such as Jetson tx and Jetson AGX Xavier, which can increase the computational power and speed required for fish detection and tracking [7],[16].

2. Dataset Properties

2.1. Context

The Southeast Area Monitoring and Assessment Program (SEAMAP) reef fish video survey has been conducted annually since 1992. Traditional fisheries gears (e.g. trawls) often damaged the habitat and were ineffective for the reef species that the National Marine Fisheries Service is required to manage. In the late 80’s underwater video became more accessible and researchers began investigating how to use the technology to evaluate abundance of marine resources. The Public Access to Research Results (PARR) initiative requires scientific data to be accessible to the public and thus PARR metadata from the survey is archived and available through the National Centers for Environmental Information [27].

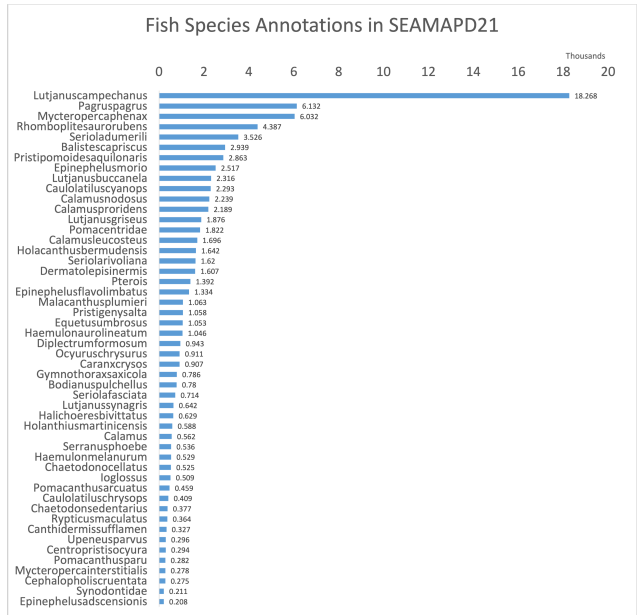


Figure 1. Number of Occurrences per Species in SEAMAPD21

The survey is primarily conducted on the high-relief shelf edge break habitat of the northern Gulf of Mexico from the U.S.-Mexico international border to the Dry Tortugas, FL. Depths range from 15 - 200m. The map is shown in Figure 2.

Composition of the reef habitat varies widely from low-relief (<1m) relic shorelines, to high-relief (>5m) hermatypic coral reef systems. The underlying research design is optimized for species in the snapper-grouper complex and therefore targets habitats they occupy. While useful for fishery stock assessment purposes, this tactic is responsible for the main challenge of this dataset: the diversification of images per species vary widely from 18,268 (red snapper, *Lutjanus campechanus*) to 3 (Coney, *Cephalopholis fulva*). The diversity of the documented species in the dataset is

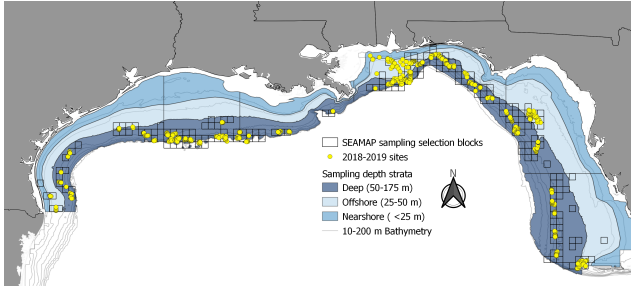


Figure 2. Survey map

directly correlated to the habitat where the cameras were deployed; *Lutjanus campechanus* make up the majority of the reef fish population captured on video because they are the survey target species. The quantitative distribution of occurrences is shown in Figure 1.

2.2. Data collection

The concept behind fishery-independent surveys is that the deployments are conducted at randomly selected locations in a predefined universe while the methods and gear remain standardized.

The primary deployment gear in the survey has been a stationary, ground-tended array with various camera technology utilized. SphereCam provides 360° field-of-view by stitching imagery from five cameras and stereo-vision in a single orientation via an attached satellite camera. Imagery is currently manually annotated from a single camera to match historic processes.

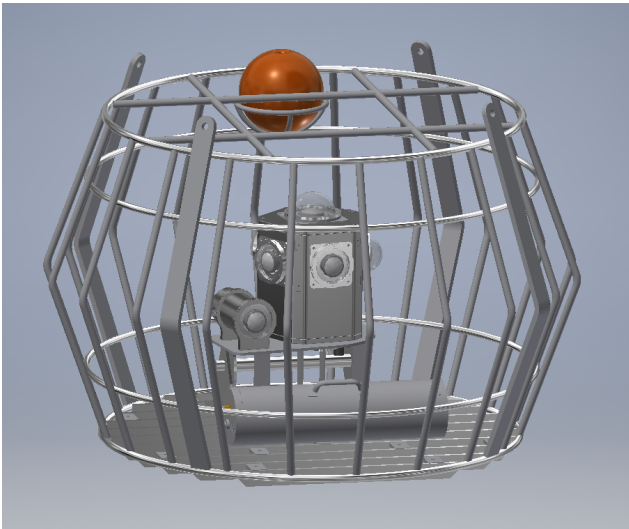


Figure 3. 3D rendering of small camera array used for video collection in the Gulf of Mexico

The camera arrays are baited and deployed for 30 minutes at each location. Today, we sample 600 sites annually. All deployments conducted over the course of a survey

are conducted concurrently with Shipboard Computing Systems (SCS) event information to enable geospatial and temporal coordination of data collection across multiple sensors (e.g. GPS, CTD, Cameras).

2.3. Annotation Process

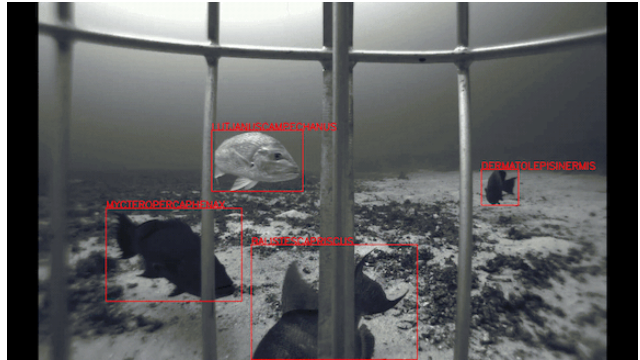


Figure 4. Initial species-level predictions on SEAMAPD21

While at sea, video is compiled over 10-15 deployments on a daily basis using onboard camera computer systems. Video from the stereo cameras are then stitched into the 360 degree imagery overnight and all associated metadata compiled (e.g. camera malfunctions). Upon return from the survey the computer array is removed from the ship and re-connected to laboratory networks to conduct backups and begin manual annotation of the fish and habitat data. In 2021 we began preliminary tests of automated identification and classification systems to evaluate the efficacy of near real-time annotation of video data at sea.

One video tape out of four possible from each station was randomly selected for viewing. Twenty minutes of the videos were viewed, starting from the time the view cleared of suspended sediment after impacting the seafloor. Viewers identified and enumerated all species that have federal management plans in place (FMP species) to the lowest taxonomic level during the 20 minute viewable segment. Species specific counts are derived at each individual site using the *MaxN* method [12]. *MaxN* is defined as the highest count observed over the course of the video for each species. *MaxN* is preferred because it avoids double counting fish that are swimming in and out of the camera field-of-view. Therefore it represents the conservative maximum number of individuals observed during a deployment.

3. Baseline Experimentation

Both fish detection and fish species classification are required to calculate an index of relative abundance from SEAMAP video data. Due to relatively low inter-class variance between fish species, we separate these two tasks of detection and classification to different networks. For detection, images are randomly split into training, testing, and

validation sets with 50% of the *Set1* annotations used for training, 20% of the *Set2* annotations used for validation, and another 20% of the *Set2* annotations used for testing. For classification, the 54 species classes with more than 10 individuals are trained on. We exclude the other species as they have too few samples for our baseline methods to train well on. The data is split by track in a 60:20:20 ratio and filtered by bounding box size and location to remove badly truncated and occluded examples. A track is defined as all occurrences of an individual fish. The models are trained using COCO pre-trained models for detection and ImageNet pre-trained models for classification. All FPS values are calculated using a Geforce RTX 2080 GPU.

We provide baseline fish detection (mAP) and species classification (accuracy) results using popular object detection and classification networks. The mAP of YOLOv4 for different IOU values is shown in Figure 5. Open source code from OpenMMLab is used for training most of our baseline models. Both the MMDetection [8] and MMClassification toolboxes provide implementations of many popular deep learning architectures in pytorch that are simple to use and modify. With OpenMMLab, researchers can easily reproduce and verify our baseline results. Our training parameters can be found at <https://github.com/SEFSC/SEAMAPD21>.

3.1. Results

Model	Backbone	mAP	mAP ₅₀	FPS
FRCNN [32]	ResNet-50 [15]	35.6	60.7	14.8
FRCNN	ResNet-101	37.7	64.7	11.0
FRCNN	ResNeXt-50 [39]	36.6	65.1	13.0
FRCNN	ResNeXt-101	36.2	60.9	13.1
RetinaNet [22]	ResNet-50	38.8	68.8	15.1
RetinaNet	ResNet-101	39.3	68.8	11.2
RetinaNet	ResNeXt-50	36.0	68.4	13.4
RetinaNet	ResNeXt-101	39.4	70.3	13.2
YOLOv3 [31]	DarkNet-53	29.1	55.5	77.8
YOLOv4 [2]	CSPDarkNet-53	33.12	63.35	55

Table 1. Mean Average Precision (mAP) results on SEAMAPD21 Fish Detection. The overall mAP is averaged over multiple Intersection over Union (IOU) thresholds, while mAP₅₀ is only calculated with an IOU threshold of 0.5.

Model	Top1 Accuracy	Top5 Accuracy	Track Accuracy
ResNet-50	69.6	90.5	70.2
Mobilenetv2 [35]	68.5	91.1	70.4
ResNeXt-50	68.8	90.6	70.1
SE-ResNet-50 [17]	72.4	92.2	73.9

Table 2. Accuracy Results on SEAMAPD21 Species Classification. Track accuracy is calculated using majority vote to predict species at a track level.

From our baseline experiments, the best fish detection

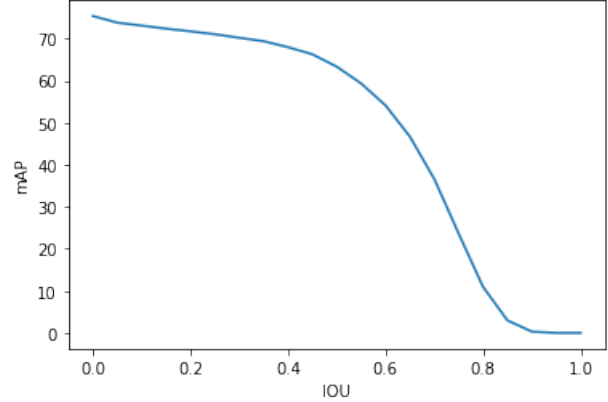


Figure 5. YOLOv4 mAP vs IOU

model is RetinaNet with a ResNeXt-101 backbone, achieving an mAP of 39.4. The best species classification model is SE-ResNet-50, with an accuracy of 72.4. While RetinaNet is the best performing model in terms of mAP, FPS can be an equally important metric for object detection networks depending on the application. For real time applications that will live on limited hardware, lightweight networks such as YOLO are preferable despite their lower mAP performance.

The SEAMAPD21 dataset presents a challenge in both the detection and classification tasks required for automated analysis. The large variance in water transparency and the imbalanced species distribution have a considerable impact on model performance. Future experiments may benefit from using GANs[40] to translate murky images to clear images similar to defogging. Using long-tail recognition techniques [23] may also increase classification accuracy for tail classes with fewer samples, increasing the number of species we can distinguish to include those with fewer than 10 individual samples.

4. Conclusion

This study demonstrated the value of comparing and contrasting various machine learning techniques through deep learning to facilitate automated analysis to count and identify fishes from fishery-independent surveys. Several approaches used in this study clearly performed better, although this dataset is only the first step on a longer journey to continue collecting annotated imagery of fish populations. Given the long-tail state of the SEAMAPD21, future researchers will require greater diversity for enhanced classification precision in the future. We believe improvement in approaches may need to be coupled with improvement in data quality to reach consistent and highly accurate results. This research is vital to reducing annotation costs and improving timeliness of survey results that are critical inputs in numerous Gulf of Mexico and South Atlantic fishery stock assessments.

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